EQUIPMENT SUBMITTAL TR

REVISE AND RESUBMIT

Transmittal No:		**Date:**	
Project:	Meridian Data Centre Campus, Phase 1	**Spec Section:**	26 33 53
Vendor:	VoltEdge Power Systems	**Revision:**	
Status:	FOR REVIEW	**EPC Lead:**	YOU Electrical

Submittal Item Description

We submit for engineering review and approval the product datasheet, compliance matrix, and test certifications for the following equipment:

Model Number	Description	Qty	Manufacturer
VoltEdge BC-200	VoltEdge BC-200 High-Rate VRLA Battery Cabinet Rack	24	VoltEdge Power Systems

Review Comments / Action Taken

Engineering Review Result: REVISE AND RESUBMIT
Comments:
- Approved with no major comments. All values check compliant.

Product Datasheet: VoltEdge BC-200

The VoltEdge BC-200 is a matching battery cabinet rack housing high-rate Valve Regulated Lead-Acid (VRLA) batteries connected in series to provide DC back-up power to the UPS inverter during utility outages.

Technical Parameter Specifications

Parameter Name	Claimed Value	Claimed Condition	Spec Limit
Nominal energy kwh	576.0 kWh	at 10h discharge rate	576.0 kWh
Load power factor	0.9	discharge condition	1.0
Dc bus voltage	400.0 V	nominal DC bus	480.0 V
Shelf life months	missing months	without recharge	6.0 months
Cabinet weight kg	3200.0 kg	fully loaded	2800.0 kg
Operating temp limit c	20.0 C	operating ambient limit	25.0 C
Cable entry conduit	bottom	installation config	top_or_bottom
Lvd contactor rating	missing	standard coil	continuous

Notes:
1. Cabinet houses 40 blocks of 12V 200Ah batteries in series (nominal DC bus 480VDC). Stated 576 kWh rating is the combined bank nominal capacity rating. 2. Backup autonomous runtime of 10 minutes is rated under a load power factor condition of 0.9 PF.

Equipment Performance Curves

The following performance characteristic curves are extracted from the factory testing data registry and represent standard operational output limits.

Performance curve chart not required for this equipment class.

* Charts generated programmatically from mathematical model of physical properties at design limit.

Clause-by-Clause Conformance Statement (Page 1)

Below is our conformance statement indicating compliance with each clause of Section 26 33 53 of the construction specifications.

Spec Clause	Parameter / Requirement	Conformance Claim	Remarks / Evidence
26 33 53 Part 2.3.2.A	BAT-CAPACITY-SHORTFALL	NC	Non-Conformant. Vendor datasheet claims nominal capacity is 550 kWh, which when converted (550,000 Wh / 480 V = 1145.8 Ah) is below the required 1200 Ah, indicating a capacity shortfall of 54.2 Ah (4.5%).
26 33 53 Part 2.3.2.A	BAT-CAPACITY-UNIT	C	Conforms. Battery capacity is quoted as '576 kWh' in the submittal. Converting this (576,000 Wh / 480 V = 1200 Ah) matches the required capacity of 1200 Ah.
26 33 53 Part 2.3.2.B	BAT-PF-CONDITION	NC	Non-Conformant. Datasheet claims 10-minute battery runtime is rated at a load power factor of 0.9 PF. The spec requires battery runtime to be rated at a load power factor of 1.0 PF (unity), which represents a larger active power draw.
26 33 53 Part 2.3.2.C	BAT-DC-BUS-VOLTAGE	NC	Non-Conformant. Battery cabinet DC bus voltage is listed as 400 VDC. The spec requires a 480 VDC nominal bus voltage to match the UPS inverter requirements.
26 33 53 Part 2.3.3	BAT-SHELF-LIFE	NC	Non-Conformant. The battery shelf life (required to be >= 6 months without charge) is missing from the battery cabinet specification sheet.
26 33 53 Part 2.3.5.A	BAT-CABINET-WEIGHT	NC	Non-Conformant. Fully loaded battery cabinet weight is listed as 3200 kg. Spec limits the maximum floor-load weight per cabinet to 2800 kg to prevent exceeding structural slab limits.
26 33 53 Part 2.3.6.B	BAT-OPERATING-TEMP	NC	Non-Conformant. Battery cabinet datasheet lists operating temperature of 0°C to 20°C. Spec requires full battery warranty and operating parameters to hold from 0°C to 25°C ambient.

Clause-by-Clause Conformance Statement (Page 2)

Spec Clause	Parameter / Requirement	Conformance Claim	Remarks / Evidence
26 33 53 Part 2.3.8	BAT-CABLE-ENTRY	**NC**	Non-Conformant. Battery cabinet is configured for bottom cable entry only. Spec requires dual entry (top or bottom cable entry conduits) to match overhead cable trays.
26 33 53 Part 2.3.9.C	BAT-LVD-CONTACTOR	**NC**	Non-Conformant. Low Voltage Disconnect (LVD) contactor rating is omitted from the battery cabinet component details.

CERTIFICATE OF CONFORMANCE

Factory Quality Assurance Department

We hereby certify that the equipment listed below has been manufactured, tested, and inspected in accordance with all active factory testing specifications and international standards.

Equipment Model: VoltEdge BC-200
Serial Numbers: VOL-2026-2601 to 08
Test Standards: IEC 62040-3, IEC 62271-200, ISO 8528-5 as applicable.
Quality Inspector: Dr. H. Bose (Director of QA)
Date of Inspection: January 8, 2026

Quality Inspector Signature

Plant Manager Signature